

Project 2: Conceptual Database Design

Due date: 11 February 2026 before 9 p.m.

In previous projects, students have already selected a real business domain of their interest and identified the requirements specification of the database application for the selected business domain. In this project, students are required to come up with a conceptual data model represented in the format of an Entity-Relationship (ER) model with respect to the requirement specification.

Things to do

Extend the previous project by coming up with a **conceptual data model** represented in the format of an Entity-Relationship diagram (ERD) with respect to the **user requirements** (*data and transaction requirements*). If the requirement specification from previous project is changed, all the changes must be listed, and the new database requirement specification must be declared. The number of entity types and that of relationship types must support the core business requirements. Attributes associated with each entity and relationship type and the candidate, the primary and the alternate keys must be identified appropriately. The user transactions and the constraints about data with respect to the user transactions must also specified appropriately. If the requirement specification from previous project is changed, all the changes must be listed, and the new database requirement specification must be submitted.

Deliverables

A **document** in portable document format (PDF) that contains the following information:

1. The **user requirement specifications** which include the data and the transaction requirement specifications. The list of database specification changes (if any).
2. The **ERD** representing the **conceptual data model** and corresponding to the user requirement specifications. UML notation is recommended; Chen and Crow's foot notations are alternatives; No other notation is allowed. The entity types, the relationship types, the attributes including keys associated with entities and relationships, and the structural constraints must be appropriately specified with respect to the business rules.
3. The **pathway** of verifying whether the conceptual data model supports all user transactions.

Submissions

Name the report file as <ProjectID>_<GID>_<last-three-digit-of-StudentID for every team member>_report.pdf, for example, "**P2_G01_999_988_977_report.pdf**".

Information and Communication Technology Program
ITCS442 Relational Database Design